

Research paper

A fair and effective approach to managing distributed energy resources through peer-to-peer energy trading with load prioritization among smart homes

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ABSTRACT

This paper proposes a novel approach to effectively manage distributed energy resources (DERs) by facilitating peer-to-peer (P2P) energy trading among smart homes. This system incentivizes peers to trade surplus energy, thereby maximizing social welfare. The proposed day-ahead demand-side management (DSM) technique considers load prioritization based on individual user needs. It employs an optimal energy allocation policy that encourages peers to purchase energy at reduced rates while selling it at advantageous prices. This strategy optimizes the allocation of traded energy to higher-priority loads, followed by lower-priority ones, all while considering user preferences. The proposed method results in an average reduction in energy costs from 14.80 % to 2.70 % for all consumers and an average increase in revenues from 17.35 % to 9.07 % for all prosumers compared to the grid and previously proposed studies. Moreover, the proposed approach significantly enhances the consumer satisfaction Index (SI), with a 46.8 % increase over the previous research study. Simulation results demonstrate substantial economic benefits for all market participants, improved SI, enhanced community self-sufficiency, and reduced dependence on the main grid. The proposed methodology offers a practical and fair solution for managing DERs through P2P energy trading by implementing load prioritization within smart homes. It highlights the potential to establish a sustainable and efficient energy ecosystem, thereby enhancing the effective utilization of surplus energy from DERs.

1. Introduction

1.1. Background and literature review

The global energy demand has increased from 86 EJ to 503 EJ, and it will rise to 1000 EJ in 2050 (Zhu et al., 2020). Many countries seek to expand generation through new plant installations or upgrades to fulfill these energy demands and rely on fossil fuels to overcome energy deficits. However, this leads to CO₂ emissions, which rose from 310 ppm in 1950 to 390 ppm in 2010 (Zhang et al., 2019) and clashes with green objectives.

To address this challenge, solar photovoltaic (PV) and wind energy are emerging as viable alternatives to fossil fuel-based power generation due to their fuel-free and zero-carbon emission advantages. Moreover, integrating these distributed energy resources (DERs) and developing a smart grid is an efficient way to fulfill energy demands (Momoh, 2009). The installation of DERs will rise to 75 % by 2030, with PV installation having the highest investment, and the residential sector will have 50 % of the PV investments (Nheide, 2020). This way, these DERs promote with time and provide business opportunities for various entities, especially evident in the popularity of community microgrids. These community microgrids are becoming increasingly popular as they offer different business opportunities to control energy consumption, increase

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Nomenclature

Z	Set of Peers.
P, C	Set of prosumers and consumers.
P2P MO	Peer-to-Peer market operator.
DERs	Distributed Energy Resources.
D_i	Demand for the energy of individual consumer.
δ	Individual consumer eagerness.
$T.S_i$	Surplus energy of all prosumers.
$T.D_i$	Actual demand for the energy of all consumers.
L_s	Reallocated load.
L_m	Modified demand for the energy of all consumers after reallocation.
SDR	Supply-to-demand ratio.

$e_{i,g}, e_{e,g}$	Energy import, and export with the main grid.
C_w	Set of consumers' willingness.
P, Q, R	Priority levels of sets of appliances.
λ	Prosumer reputation factor.
A_n, B_n, C_n	Sets of priority A, B, and C appliances.
$E_{A_n,t}, E_{B_n,t}, E_{C_n,t}$	Energy demand of highest, medium, and least priority load.
R_p	Reference pricing signal.
$c.p_i, e.p_i$	Set of Offered and proposed prices of consumers and prosumers.
$x_{e,p2p}, x_{e,g}$	Export price to the P2P market and grid.
$x_{i,p2p}, x_{i,g}$	Import price from the P2P market and grid.
SI	Satisfaction Index.

efficiency, and adapt energy sources to local needs using DERs and load control at the household level (Zhang et al., 2020).

Under such circumstances, consumers can become prosumers by flexibly supplying surplus energy to their neighbors or the grid through P2P energy trading, attracting great interest in promoting local energy trading and balancing among developing prosumers (Chen and Chang, 2018; Dimeas, 2014). Due to advancements in information and communication technologies (ICTs), this shift towards decentralized networks marks a transition from centralized to distributed energy networks.

The recent growth in adopting DERs installations is significant but does not meet the huge energy demand (Electricity Market Report, 2021). Therefore, residential participants can implement demand response (DR) programs to efficiently utilize distributed generation and reduce energy costs among residential participants in community microgrids (Fahrioglu et al., 2012). Proactive consumers and prosumers are driving this shift by using energy storage systems (ESS), flexible loads, generation from DERs, and smart meters to manage their resources under demand response (DR) programs (Parag and Sovacool, 2016; Qiu et al., 2018). A study developed a distributed method for residential transactive energy systems to optimize the energy management of HVAC systems and batteries to reduce the total energy cost (Yang and Wang, 2021).

Considering the energy availability from DERs, residential consumers can further minimize their energy costs. Consumers can modify their energy demand pattern with respect to the available energy from PV, wind, and ESS (Tavakkoli et al., 2021; Gholami and Sanjari, 2021; Slama, 2021; Hlalele et al., 2021). The authors in (Tavakkoli et al., 2021) utilized the Stackelberg game model to meet the energy demands of HVAC loads through energy generated by PV and wind sources. Moreover, the scheduling of appliances is performed considering the PV energy and load elasticity (Gholami and Sanjari, 2021). Additionally, the scheduling scheme incorporated EVs in conjunction with PV and ESS (Slama, 2021). In (Hlalele et al., 2021), researchers extended the appliance scheduling to integrate wind power as well as the charging and discharging of EVs. A proposed mechanism in (Aznavi et al., 2020) enables P2P energy trading among EVs equipped with solar panels, aiming to reduce total costs.

Many researchers have worked on P2P energy trading among smart homes to diminish reliance on the main grid during various time intervals when the energy from DERs is available. In a P2P setting, prosumers are the entities that can both produce and consume energy within their premises. On the other hand, consumers only need energy to run their electrical appliances. Smart homes can further reduce their energy costs by engaging in P2P energy trading and purchasing energy from neighboring prosumers (Soto et al., 2021). A research study in (Korjani et al., 2022) incorporates PV and ESS within prosumers' premises to establish P2P energy trading. It works on an offline clustering

for effective energy management and achieves self-sufficiency as a priority rather than engaging in trading. Another cost minimization problem is presented in (Mehdinejad et al., 2022) that enables P2P energy trading between prosumers, retailers, and consumers in a decentralized fashion. Clusters having prosumers and consumers with the same energy load profiles are grouped in (Luo et al., 2022) to facilitate P2P trading. It operates on a self-sufficient rate, prioritizing allocating maximum energy to prosumers to enhance self-sufficiency instead of energy trading. Additionally, a grid-connected prosumer-centric residential smart community employs cooperative game theory to exchange, regulate, and schedule excess energy generated by DERs with other neighboring prosumers. This approach aims to increase system efficiency and reduce carbon emissions. (Gbadega and Sun, 2022).

The energy trading cost at each time interval of the trading period is an essential factor influencing consumers' energy costs and prosumers' revenue. Authors in (Zheng et al., 2022) have used fixed costs for P2P trading. It reduces energy bills for the consumers, but bidding appropriate prices by the peers can further minimize it. Another research study in (Pena-Bello et al., 2021) has calculated the trading price by considering the energy availability from the PV and feed-in-tariff (FiT) as a lower limit and the grid price as the upper limit. A research study has performed P2P energy trading based on uniform pricing among the market participants (Amin, 2020). Authors in (Morstyn and McCulloch, 2019; Yao et al., 2022) have taken wholesale prices for the P2P trading among the prosumers only, and no consumers are involved. Another study in (Anoh et al., 2020), chooses the highest price proposed by the prosumer among multiple prosumers.

Similarly, in (Fernández-Blanco et al., 2014), the prosumer with the highest selling price exercises control over the overall selling price within the P2P market. In both cases, prosumers determine the trading price, yet the primary limitation lies in the absence of consumer engagement in the trading process. The research studies in (An et al., 2020; Zhao et al., 2022) select the trading prices in such a way that is in favor of prosumers, but consumers may get affected because of not getting the required energy at a reasonable price. The trading cost is computed based on SDR value and buy and sell bidding values (Neves et al., 2020), but it does not consider the grid tariff, and there might be some intervals where the trading cost is more than the grid price, which is not feasible. Also, in (Park et al., 2021), the calculated trading price lies between Time of Use (ToU) and FiT.

In recent times, the implementation of DSM and DR programs in the P2P setting has gained significant attention, enabling the effective utilization of green energy to diminish dependence on the main grid (Liu et al., 2020; Görgülü et al., 2022; Rodrigues et al., 2020). Authors in (Liu et al., 2020) have implemented residential DR in the P2P setting, taking only a PV source as a generating source and a few appliances, i.e., electric pumps, refrigerators, and ESS. In (Görgülü et al., 2022), the authors implemented the DR program in P2P trading without

considering appliance scheduling, which can lead to consumer dissatisfaction. In (Rodrigues et al., 2020), the authors introduced P2P trading involving PV and ESS, where they assumed the prices and omitted the consideration of smart home appliances within the P2P community. A research study in (Kanakadhurga and Prabakaran, 2021) proposed a DR-based P2P energy trading between peers among smart homes to reduce the cost and burden on the main grid through particle swarm optimization.

It is worth noting that controlling user demand from the load side has been a topic of increasing interest within the scientific community and among researchers (Mortaji et al., 2017; Pipattanasomporn et al., 2014; Logenthiran et al., 2012). A dynamic, real-time energy pricing method that considers the demand for fairly distributed resources for the EV charging process is proposed in (Aljohani et al., 2021). This system reflects instantaneous energy prices in response to the microgrid's real-time energy supply and follows the inverse demand function.

However, as mentioned above, a significant portion of the literature focused on implementing DR programs in smart homes or bidding and decision-making strategies for P2P trading in the P2P market separately. In contrast, our proposed model adopts a comprehensive approach by considering surplus energy availability, consumers' energy demand, and smart appliance prioritization. The main objective of this study is to utilize the available green energy effectively by prioritizing higher-priority loads followed by low-priority loads. This approach aims to reduce reliance on the main grid and maximize the social welfare of participants without compromising on SI, which has not been considered together in the existing literature. The goal is to optimize energy utilization from P2P trading for higher-priority loads rather than relying on the main grid.

- A significant portion of the existing literature has not simultaneously considered the implementation of DSM, load prioritization, and P2P trading among smart homes.
- Within the literature, researchers have primarily focused on smart homes with a limited number of appliances, which typically include thermal or electrical loads.
- Previous studies have neglected to investigate the combined impact of load prioritization and control strategies within smart homes in P2P communities. This oversight disregards the participants' objective of maximizing social welfare.
- Only a few researchers have concentrated on implementing DR in P2P trading. Their main focus has been decision-making strategies in the P2P market for energy trading or cost minimization, often without considering the consumers' SI.

1.2. Specific contribution

In light of the literature discussed above, there is a need for an optimal energy allocation policy. This policy should consider the erratic nature of DERs and prioritize the energy needs of participants who intend to engage in P2P trading. These participants should be given priority and supplied energy from the P2P market. During time intervals when the P2P market cannot provide energy to all consumers, it is essential to prioritize load and provide energy to them while maintaining their SI.

To meet these objectives, the main contributions of this paper are as follows:

- An optimal energy allocation policy is proposed to maximize the trading time of energy for higher-priority loads, followed by lower-priority loads, all while considering user preferences.
- Appliances prioritization is performed based on available energy and demand, considering consumers' SI to optimally use the available energy from DERs. Excess energy generated by prosumers, beyond their self-consumption needs, can be sold to other consumers through P2P trading.

- A load control and reallocation mechanism are designed, following an exhaustive search algorithm to prioritize higher-priority loads over lower-priority ones. This approach aims to increase the energy hours supplied through the P2P market to the higher priority loads.
- The framework accommodates the stochastic nature of demand and supply in the P2P energy market. It ensures a dependable and efficient consumer energy supply, fosters mutually beneficial interactions, enhances consumers' SI, and maximizes overall social welfare.

1.3. Paper organization

The remainder of the paper is organized as follows: Section II presents the P2P system model, followed by the energy allocation policy in Section III. Section IV explains the smart home load control and prioritization scheme. The simulation results and findings are presented in Section V, and Section VI concludes the paper with future recommendations.

2. Peer-to-peer system model

In the proposed market, $\mathbb{Z}$ peers interact at any time, including consumers and prosumers. We have different peers who want to trade energy in the P2P market represented as $\mathbb{Z} = P \cup C$, and $P \cap C = \phi$, where P is set of prosumers and $P = \{P_1, P_2, \dots, P_n\}$. Likewise, a set of consumers C and $C = \{C_1, C_2, \dots, C_n\}$. In the P2P market, the P2P market operator (P2P MO) monitors and regulates all trading operations within a community. In the P2P energy trading market, peers with common interests and goals in energy trading and positive utility participate. A consumer informs the system of its electricity demand, D_i at a time t , along with a willingness factor C_w . This willingness factor allows the P2P MO to distinguish between consumers who wish to participate in the P2P market and those who want to purchase energy directly from the grid. In general, we can categorize consumers into two types: price-conscious and energy-conscious consumers. Price-conscious consumers focus on the cost of energy. In contrast, energy-conscious consumers emphasize the accessibility of energy, even if they have to pay a slightly higher price. Based on the consumers' willingness, the P2P MO treats it as a decision variable to categorize the energy requirements of consumers in the P2P energy market. The consumers tell the P2P MO of electricity requirements $T.D_i$ along with an eagerness such that $\delta \in C_w$. Here, $T.D_i$ indicates the overall energy demand of all consumers, wherein $T.D_i = E_{j(T)} + E_{-j(T)}$. Where $E_{j(T)}$ represents the total energy demand of the j th household throughout the day, while $E_{-j(T)}$ indicates the energy demand of households other than j , respectively.

$$T.D_i = E_{j(T)} + E_{-j(T)} = \sum_{t=1}^T (E_{A_{n,t}} + E_{B_{n,t}} + E_{C_{n,t}})_j + \sum_{t=1}^T (E_{A_{n,t}} + E_{B_{n,t}} + E_{C_{n,t}})_{-j} \quad (1)$$

Here $E_{A_{n,t}}$, $E_{B_{n,t}}$ and $E_{C_{n,t}}$ denotes the highest priority load, the medium priority load, and the least priority load, along with the willingness factor C_w with different thresholds for the different priority loads. These thresholds are input parameters of smart homes for load control. Whenever there is a mismatch between supply and demand, the central control unit monitors the continuous track of available energy in the P2P market. It sends the signal to the remote control unit to run the essential load from the P2P market or the main grid based on the willingness of the peers and the load prioritization data. All prosumers inform the system about the surplus energy $T.S_i$ and the willingness factor P_w . In addition, P_w assists the system in classifying and managing prosumers.

A function for the proposed system can be written as follows:

$$\min : f : f_a(T.S_i, T.D_i, C_w, P_w) + f_b(e_{i,g}, e_{e,g}) \quad (2)$$

$$\sum E_R = 0 \forall C_n, C \in C \quad (3)$$

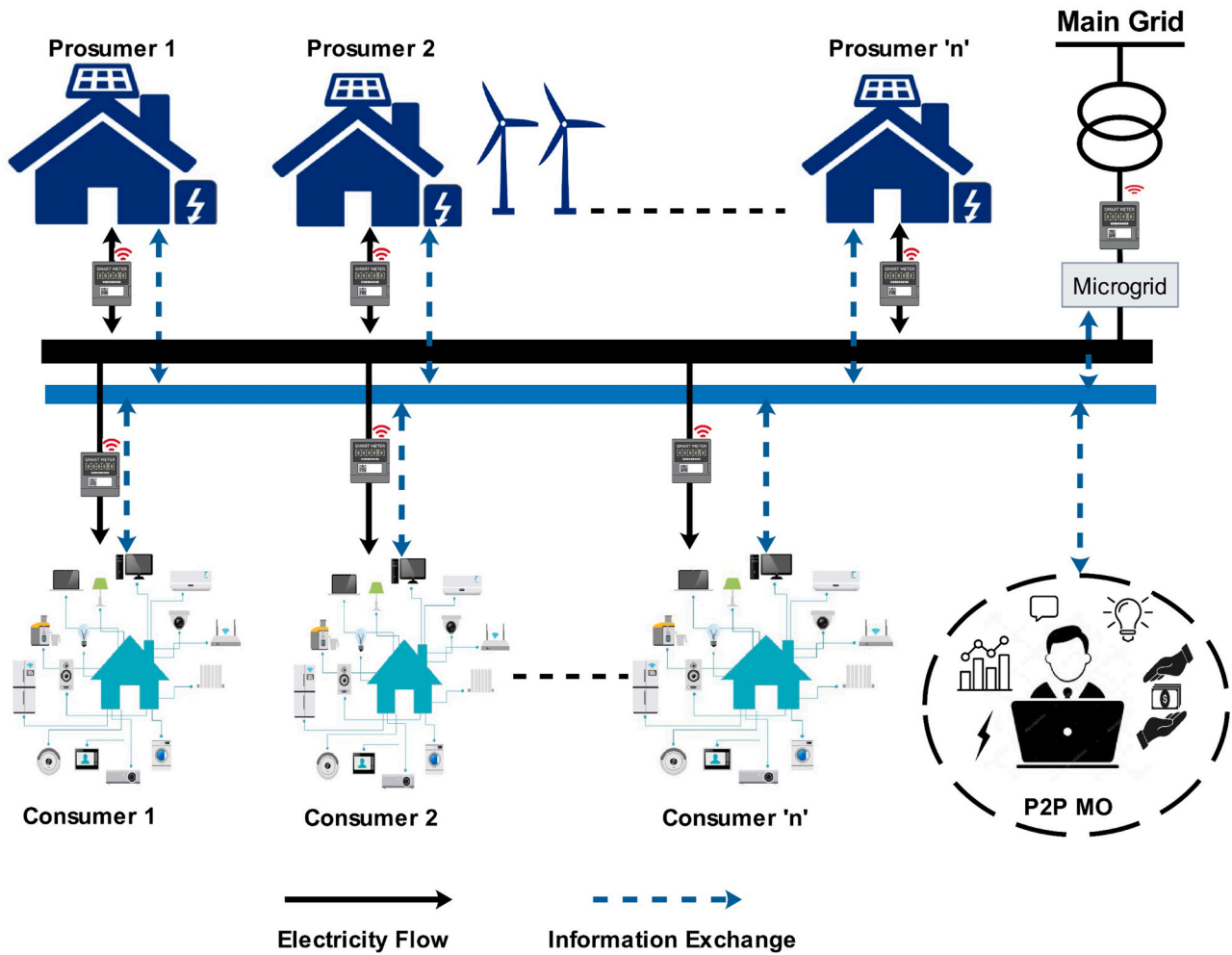


Fig. 1. An overview of a P2P network.

$$\sum C_w = e_{i,g} \quad (4)$$

$$\sum P_w = e_{e,g} \quad (5)$$

The first part of Eq. (2) represents the sets of decision variables for the P2P market, and the second part describes the energy import and export with the main grid. $T.S_i$, $T.D_i$, C_w , and P_w denotes total surplus energy, the energy required by consumers, consumers' willingness, and prosumer willingness, respectively. On the other hand, $e_{i,g}$, $e_{e,g}$ denotes the energy import and export with the main grid. The potential of the P2P energy market for local energy trading has become increasingly important. It includes price-based DR programs and optimal decision-making for participants to monitor energy supply and demand to enable better use of renewable energy. However, a challenge arises from the constant fluctuations in DERs energy generation. This production is weather-dependent and depends on factors such as sunlight and wind to generate energy from solar panels and wind turbines.

To address this issue, categorizing consumer loads in P2P communities becomes essential for optimizing the utilization of surplus energy generated by DERs. Fig. 1 illustrates a visual representation of the proposed model.

2.1. Prosumer model

After self-consumption, each prosumer P_n earns some revenue R at a time t after selling surplus energy e_s . It is a well-known fact that prosumers can earn more by exporting energy to the P2P market than by

selling it into the grid (Amin et al., 2020; Tushar et al., 2018), as demonstrated:

$$x_{e,p2p} > x_{e,g} \quad (6)$$

Any prosumer who sells surplus energy to the P2P market at a price $x_1, x_2, \dots, x_n, x_n \in x_{e,p2p}$, and earns a profit $p_1, p_2, \dots, p_n$. The total revenue earned by this prosumer at the time t is given by:

$$\max \sum_{t=1}^T R = \max \sum_{t=1}^T \{ (e_s \times x_{e,p2p} \times u(t)) + (e_s \times x_{e,g} \times v(t)) \} \quad (7)$$

Where $x_{e,p2p}$, $x_{e,g}$ denotes prosumer export prices to the P2P market and grid. On the other hand, $u(t)$ and $v(t)$ represent a binary variable that shows this prosumer is selling to the P2P market or grid at time t based on which it earns revenue.

2.2. Consumer model

Consumers' energy demand varies at different time intervals. Surplus energy in the P2P market, low prices, and consumers' willingness to buy energy from the P2P market or the grid collectively influence it. The energy price in the P2P market is generally lower than the grid (Amin et al., 2020; Tushar et al., 2018), which makes it a preferred option for consumers. However, the erratic behavior of DERs can sometimes result in low energy generation, which is inconvenient for consumers.

Consumers' load is divided into three parts based on their daily activities to mitigate this problem. We use a willingness factor to define energy-related objectives for different consumers. As the number of

active consumers increases, it becomes critical to classify them based on various inputs. While most consumers prioritize cost savings, some may prioritize energy supply and climate comfort. Financially unstable consumers prefer money savings, while environmentally conscious consumers prefer green energy over money and comfort. Some consumers may refuse to compromise on these issues, insisting on energy supply at any cost and refusing to change their load pattern (Jhala et al., 2019). To address this, we utilize consumer willingness to optimize and efficiently use distributed generation based on their needs and load patterns. We study consumer demand behavior in the energy market. A consumer requires energy e_r at a given time t and pays the price after trading. However, buying energy from the P2P market can be cheaper than buying it from the grid (Amin et al., 2020; Tushar et al., 2018) and can be written as follows:

$$x_{i,p2p} < x_{i,g} \quad (8)$$

Where $x_{i,p2p}$ shows import prices from the P2P market and $x_{i,g}$ shows import price from the grid. Thus, most consumers are highly willing to buy from the P2P market. To quantify this willingness, we use eagerness, δ , which ranges between 0 ~ 1, such that $\delta \in C_w$. We define thresholds based on the eagerness to distinguish consumers based on their demand. Consumers with a higher eagerness factor ($\delta \geq 0.8$) are willing to reallocate a load of class B and C appliances from energy deficits to surplus intervals to buy energy at lower prices in the P2P market. Conversely, consumers with lower eagerness ($0.3 < \delta < 0.8$) need energy for Class A and B appliances and are only willing to reallocate their class C appliances. Some consumers with low eagerness ($\delta \leq 0.3$) are unwilling to adjust their load pattern and need energy from the P2P market or the grid at any price. The proposed study aims to maximize the hours of energy supplied to the highest priority loads in the P2P market by reallocating the supply connection based on load priority. Eq. (9) calculates the ratio of energy served to consumers from the P2P market. This ratio considers the time intervals when energy is supplied as $E.P_{(T.intervals)}$, and the time intervals when energy is required as $E.D_{(T.intervals)}$, along with the total number of appliances (N) and the appliances in a specific priority level (n).

It presents the supply-to-demand ratio (SDR) in a P2P energy market at any time t . If $SDR > 1$, it indicates a surplus energy interval in which the available surplus energy from the prosumers exceeds the demand from the consumers. Conversely, if the $SDR < 1$, it shows an energy deficit interval where the demand for energy from the P2P market exceeds the available surplus. It is important to note that load priority can vary depending on the user. However, the ultimate goal is to optimize the energy hours served to a higher priority load, considering consumers' rationality regarding energy demand at lower prices. The priority levels of three sets of appliances, class A, class B, and class C, are determined by analyzing their consumption at each interval of time per day, denoted as t_1 , t_2 , and t_3 , and the values of P , Q , and R are determined using Eqs. 10, 11, and 12 respectively. By reallocating the supply connection based on load priority, this strategy can help ensure that the highest priority loads receive the energy they need while optimizing the use of available energy resources.

$$SDR = \sum_{n=1}^N \frac{E.P_{(T.intervals)}(n)}{E.D_{(T.intervals)}(n)} \quad (9)$$

$$P = \frac{t_1}{t_1 + t_2 + t_3} \quad (10)$$

$$Q = \frac{t_2}{t_1 + t_2 + t_3} \quad (11)$$

$$R = \frac{t_3}{t_1 + t_2 + t_3} \quad (12)$$

Here, P , Q , and R define the priority set of prioritized loads of classes A, B, and C, respectively. As consumption increases, the energy costs for

consumers also increase. To optimize overall social welfare, consumers need to strike a balance among different priority loads, which will help lower their energy costs and allow adjustments to their load patterns. Thus, for each consumer, P , Q , and R represent the energy consumption set of priority A, B, and C loads with certain limits. Depending on the value of δ , the base load varies among consumers and must always be met. On the other hand, the lower priority load can be reallocated, and we can provide energy from the P2P market during the reallocated time intervals for specific consumers. As a result, the constraints for the lower and upper limits of each priority load can be defined as follows:

$$P_n^{\min} \leq P_n \leq P_n^{\max} \quad (13)$$

$$Q_n^{\min} \leq Q_n \leq Q_n^{\max} \quad (14)$$

$$R_n^{\min} \leq R_n \leq R_n^{\max} \quad (15)$$

$$s.t., T.D_i, T.S_i, \delta(t) \quad (16)$$

Any consumer who buys required energy e_r from P2P, $e_{i,g}$ from the grid and e_r' from the P2P market at t_{rs} , where $x_{1,p2p}, x_{2,p2p}, \dots, x_{n,p2p}, x_n \in x_{i,p2p}$, and $x_{1,g}, x_{2,g}, \dots, x_{n,g}, x_n \in x_{i,g}$ represents the prices per kWh from P2P and grid, and this consumer receives a bill; then the total energy cost of this consumer is given by:

$$\min \text{ Bill}_i = \sum_{t=t_{rs}}^{48} \left\{ \begin{array}{l} \&(e_r \times x_{i,p2p} \times a(t)) + \\ \&(e_r' \times x_{i,p2p} \times b(t_{rs})) + \\ \&(e_{i,g} \times x_{i,g} \times c(t)) \end{array} \right\} \quad (17)$$

$$s.t., T.D_i, T.S_i, \delta(t) \quad (18)$$

Where e_r denotes energy bought from P2P e_r' is the energy given to this consumer for low-priority loads (class B or C) at reallocated interval t_{rs} , and $e_{i,g}$ the grid with respective import prices. Moreover, $a(t)$, $b(t_{rs})$, and $c(t)$ are binary decision variables to denote the status of consumers buying energy from the P2P market at the current and reallocated time intervals and grid.

3. Energy allocation policy

This section summarizes the energy allocation process among consumers in the P2P market and the mechanism for reallocating energy in case of a demand-supply mismatch. It describes how consumers who have reported a high willingness to reallocate their low-priority load are allocated energy from the P2P market at the reallocated time interval t_{rs} .

The P2P MO initiates the market by broadcasting a reference market pricing signal R_p to all peers. Consumers submit information about their energy demand, offered prices, and eagerness as $\{D_i, c.p_i, \delta_i, D_{i+1}, c.p_{i+1}, \delta_{i+1}, D_{i+2}, c.p_{i+2}, \delta_{i+2}, \dots, D_n, c.p_n, \delta_n\}$ while prosumers submit information about surplus energy and proposed prices as $\{e.s_i, e.p_i, e.s_{i+1}, e.p_{i+1}, e.s_{i+2}, e.p_{i+2}, \dots, e.s_n, e.p_n\}$. The trading of energy in P2P markets involves the participation of both consumers and prosumers who submit their respective offered and proposed prices to the P2O MO. However, both parties must follow the following criteria (Tushar et al., 2018).

$$e.p_i \geq R_p \quad (19)$$

$$c.p_i \leq R_p \quad (20)$$

Where $e.p_i$ shows the price proposed by the prosumer, and $c.p_i$ shows the offered price by the consumers at which they want to trade. This allocation mechanism ensures that the P2P market remains competitive and efficient. By broadcasting the R_p , all peers have access to the same information. Furthermore, using proposed and offered prices ensures that the energy trading is based on a mutual agreement between the prosumers and consumers. Consumers are sorted in descending order based on δ to buy and reallocate their low-priority load. In cases where multiple consumers have submitted the same δ , the consumer with the

higher offered price will be declared the winner. Prosumers are arranged in descending order based on their prosumer reputation factor λ , which is calculated as the ratio of surplus energy to the proposed price submitted to the market and can be written as:

$$\lambda_i = \frac{e.s_i}{e.p_i} \quad (21)$$

Algorithm 1. Energy allocation among peers.

interval, as shown below:

$$SI = \frac{(E_{allocated})_{n,p2p}}{D_{i,p2p}} \quad (23)$$

4. Smart home load control scheme

The proposed framework for implementing P2P energy trading between smart homes consists of two control modes: central and remote control units. The remote control unit, an IoT-based smart meter

Inputs:

P2P MO broadcasts the reference pricing signal to the peers

$\{D_i, c, p_i, \delta_i\} = \{D_1, c, p_1, \delta_1, D_2, c, p_2, \delta_2, \dots, D_n, c, p_n, \delta_n\}$ % Consumers' demand, willingness, and offered prices

$\{e.s_i, e.p_i\} = \{e.s_1, e.p_1, e.s_2, e.p_2, \dots, e.s_n, e.p_n\}$ %Prosumers' surplus energy, proposed prices to calculate λ

1. Sort consumers based on δ and prosumers based on λ in descending order
 2. *if* $\sum_{i=1}^n T.D_i \leq \sum_{i=1}^n T.S_i$ %($T.D_i$ is total Demand in P2P; $T.S_i$ is available surplus)
 - a. For each prosumer P_i :
 - i. *If* $P_i \geq C_i$: %(P_i have more energy as compared to C_i demand)
 - Send a trading request to P_i and determine the pairwise trading price among this pair
 - Update remaining surplus at ' t '
 3. *else*:
 - Send a trading request to P_i and P_{i+1} % P_{i+1} is the next prosumer in the list, determine the pairwise trading price
 - Update remaining surplus at ' t '
 4. *if* $\sum_{i=1}^n T.D_i > \sum_{i=1}^n T.S_i$ % ($T.D_i$ is total Demand in P2P; $T.S_i$ is available surplus)
 - a. Calculate the prioritized load w.r.t submitted δ using algorithm 2
 - b. Update total demand $T.D_i$
 - c. For each prosumer P_i :
 - i. *If* $P_i \geq C_i$:
 - Send a trading request to P_i and determine the pairwise trading price among this pair
 - d. Update remaining surplus at ' t '
 5. *else*:
 - Send a trading request to P_i and P_{i+1} % P_{i+1} is the next prosumer in the list, determine the pairwise trading price
 - Update remaining surplus at ' t '
 6. Repeat steps 1 to 5 until the total demand or total surplus becomes zero
 7. *else*:
 - a. Include the main grid either as a producer or a consumer, respectively, at grid rates
- end

This factor is important because small prosumers with low surplus energy are at risk of market exit due to low surplus energy. However, proposing a low price can help them remain competitive. In such cases, prosumers with a high λ will be declared the winner, even if they have relatively low surplus energy. These two lists of sorted prosumers and consumers are now ready to trade energy with each other. The trading price at a particular time interval, t , is determined by taking the average of the trading pairs' proposed and offered prices, denoted as T_p . [Algorithm 1](#) provides details of the energy allocation process in a P2P market.

$$T_{p(t)} = \text{Average } (e.p_i, c.p_i)_t \quad (22)$$

Furthermore, consumers' SI is an essential factor that affects and motivates them to participate in P2P energy trading ([Amin, 2022](#)). It assesses the consumers' satisfaction after they trade energy from the neighboring prosumers in the P2P market, and it ranges from 0 to 1. Mathematically, we can define it as the ratio of energy allocated to a consumer to the energy required by the consumer at a specific time

installed in each smart home participating in the P2P market, is responsible for bidirectional information and load control. The remote-control unit monitors load characteristics through bidirectional communication and sends dynamic data to the central control unit. Load control is the remote control unit's primary objective, which prioritizes individual households' load based on their preferences and willingness to buy energy from the P2P market. It receives control commands from the central control unit and uses the load prioritization technique to ensure continuous energy supply to the load with the highest priority.

On the other hand, the central control unit located at the point of generation facilitates the exchange of information between participants in the P2P market. It sends price signals, demand, and surplus data and facilitates buying and selling electricity to the community microgrid through the P2P MO. It also employs the load reallocation technique, ensuring that other loads are reallocated to time intervals with surplus energy, and conducts energy trading. The most commonly used communication protocols are GSM, CDMA, and IoT. However, this study

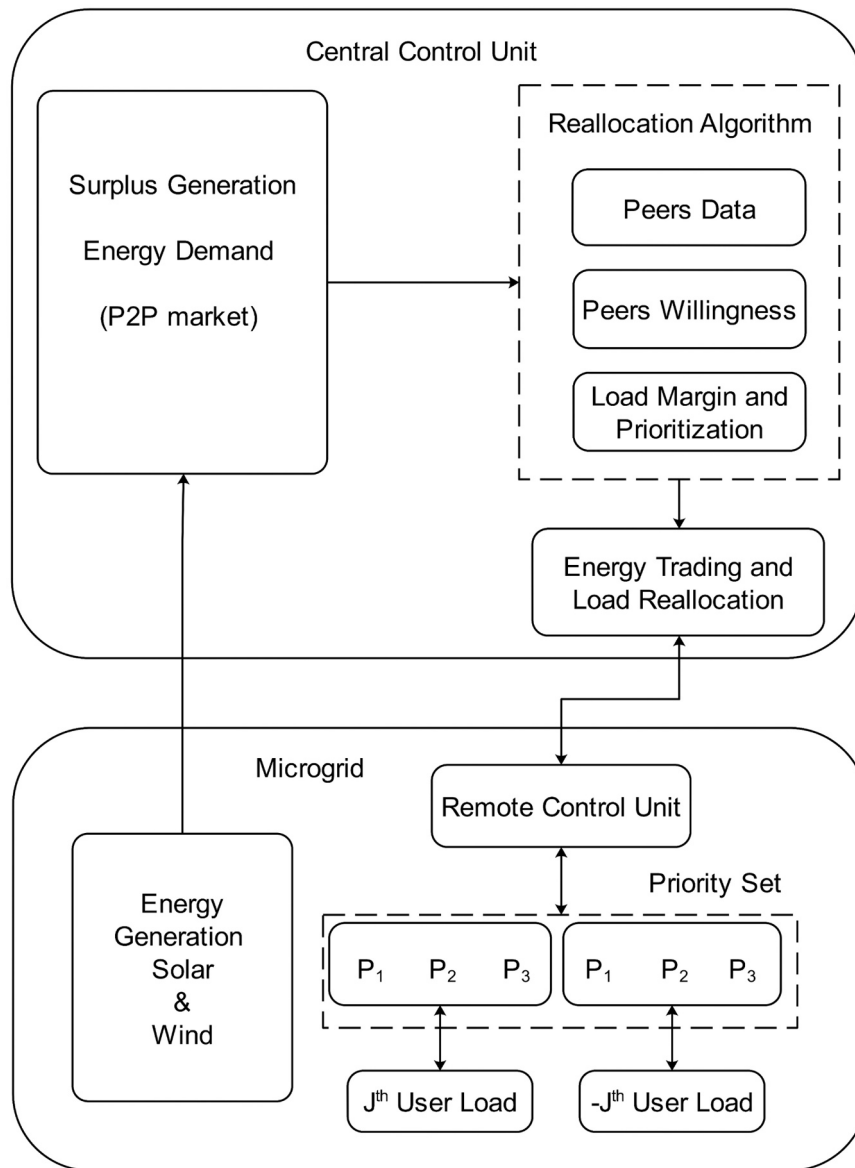


Fig. 2. Smart home load control workflow.

does not address communication reliability and availability. Fig. 2 illustrates the two-way communication between the proposed system's remote and central control units. The central control unit sends a signal indicating the load margin. In contrast, the remote control unit sends a signal with prioritized load data for load control.

5. Strategic load growth and peak clipping

The actual load curve $T.D_i$ represents the combined demand of all consumers at a given time t . Consumers are categorized based on their energy demand and δ to participate in the P2P market to reduce energy costs. Consumers more willing to participate in the P2P market can shift low-priority loads to other time intervals. This shift results in a load reallocation, L_s , from the current interval of time t to a new interval, t_{rs} . The reallocation results in a load clipping at time interval t and load growth at the new time interval, t_{rs} . The surplus energy available at t_{rs} can be provided to incentivize consumers. As shown in Fig. 2, the remote control has three supply points that can be adjusted to achieve an optimal load for reallocation. Algorithm 2 is used to determine these optimal points, which follow an exhaustive search algorithm that considers the priority-based data received from the remote control unit,

including P_1 , P_2 , and P_3 . J and $-J$ represents the J th household load and all loads except the J th household load, respectively. The mathematical function can be given by:

$$\min : f_1 : \max(T.D_i) \quad (24)$$

$$s.t \quad T.D_i > T.S_i \quad (25)$$

Eq. 24 shows the function, which is the maximum peak of the load curve, such that the total demand of the P2P market is greater than the available surplus, which must be minimized, as shown in Eq. 25. To solve the minimization problem, a function is formed in Eq. 26 to obtain the updated load curve after reallocation for the consumers who want to buy energy from the P2P market at minimized prices to reduce the total cost. For this, the formulated function can be written as:

$$\min : f : [|Load \quad Margin| - |L_h|] \quad (26)$$

Where L_h shows the load clipped or added to a specific time interval after reallocation. The load margin is the cumulative load profile of the group of consumers willing to shift their energy consumption to surplus energy time intervals t_{rs} in exchange for cheaper energy from the P2P market. It

analyzes the total load and available surplus energy in the P2P market that can be reallocated to other time intervals based on the δ of the consumers.

$$(\text{Load Margin})_t = T.D_t - T.S_t \quad (27)$$

Algorithm 2. Reallocation Algorithm.

$$\sum_{t=1}^T (L_m \times x_{p2p}) \quad (29)$$

Eq. 27 determines the load profile, i.e., the load that can be reallocated from energy deficit to surplus energy time intervals based on consumer δ . If, for a time interval t , $T.D_t > T.S_t$, then this time interval is considered overloaded. The P2P market cannot supply energy to all consumers, and this interval becomes the energy deficit interval.

```

Input Data  $\{T, D_t, T, S_t, \delta_n; \delta_n \in [0,1]\}$ 
 $t = 1$ 
converged = false
while ( $t \leq 48$ ) and not converged do           % Iterate over 48 intervals of time
    for each smart home's meter,  $i$  do
         $P_i = \text{get\_surplus\_energy}(P)$            %  $P_i$  is the surplus energy of Pth prosumer
         $C_i = \text{get\_consumer\_load}(i)$            %  $C_i$  is the Load of ith consumer
        priority = determine_p priority ( $C_i, \delta_n$ ) % Determine consumer priority
        allocation = allocate_surplus_energy( $P_i$ ) % Allocate surplus energy from the P2P market
        if  $T, D_t > T, S_t$                          % Total demand at  $t$  in P2P market > Total available surplus energy
            if (priority == "A") && (surplus_energy_interval_available > 0)
                Energy trade for priority A load; Priority B and C load should be cut off and wait for the reallocation
                 $C_i = C_i - \text{allocation}$ 
                 $P_i = P_i - \text{allocation}$ 
                surplus_energy_interval_available = surplus_energy_interval_available - 1
            else if (priority == "B") && (surplus_energy_interval_available > 0)
                Energy trade for priority A and B load and C load should be cut off and wait for the reallocation
                 $C_i = C_i - \text{allocation}$ 
                 $P_i = P_i - \text{allocation}$ 
                surplus_energy_interval_available = surplus_energy_interval_available - 1
            elseif (priority == null)
                Energy trade for all priority loads from the grid
                 $C_i = \text{get\_consumer\_load\_from\_grid}(i)$ 
            end
        else
             $P_i = P_i - \text{allocation}$ 
        end
    end
    update_ ( $T, D_t, T, S_t$ )           % update demand_and_surplus
    reallocate the  $C_i$  at surplus energy interval and energy trade at  $t_{rs}$ 
    if priorities_converged() then
        converged = true                % until no smart home's meter changes its priority
    end
     $t = t + 1$ 
end
end

```

Where $T.D_t$ is the demand of all consumers and $T.S_t$ is the total surplus energy available in the P2P market at the time t . It describes the marginal load that must be reallocated to other time intervals where the surplus energy exceeds the total demand of consumers who intend to purchase energy in the P2P market. Based on this information, it can be written as follows:

$$(\text{Load Margin})_t = \begin{cases} < 0 & \text{if } T.D_t < T.S_t \\ \geq 0 & \text{else} \end{cases} \quad (28)$$

Wherever the $T.D_t < T.S_t$ it becomes an energy surplus interval. Part of the load from overloaded time intervals can be reallocated here, considering the constraint of the consumers' δ . Therefore, reallocation is performed at each time interval, considering the load margin, to enable as many consumers in the P2P market as possible to purchase energy for higher priority loads, incentivizing their participation in the P2P market. L_m denotes the modified load curve for consumers who buy energy from the P2P market. Fig. 3 shows the data of smart home appliances before and after applying prioritization.

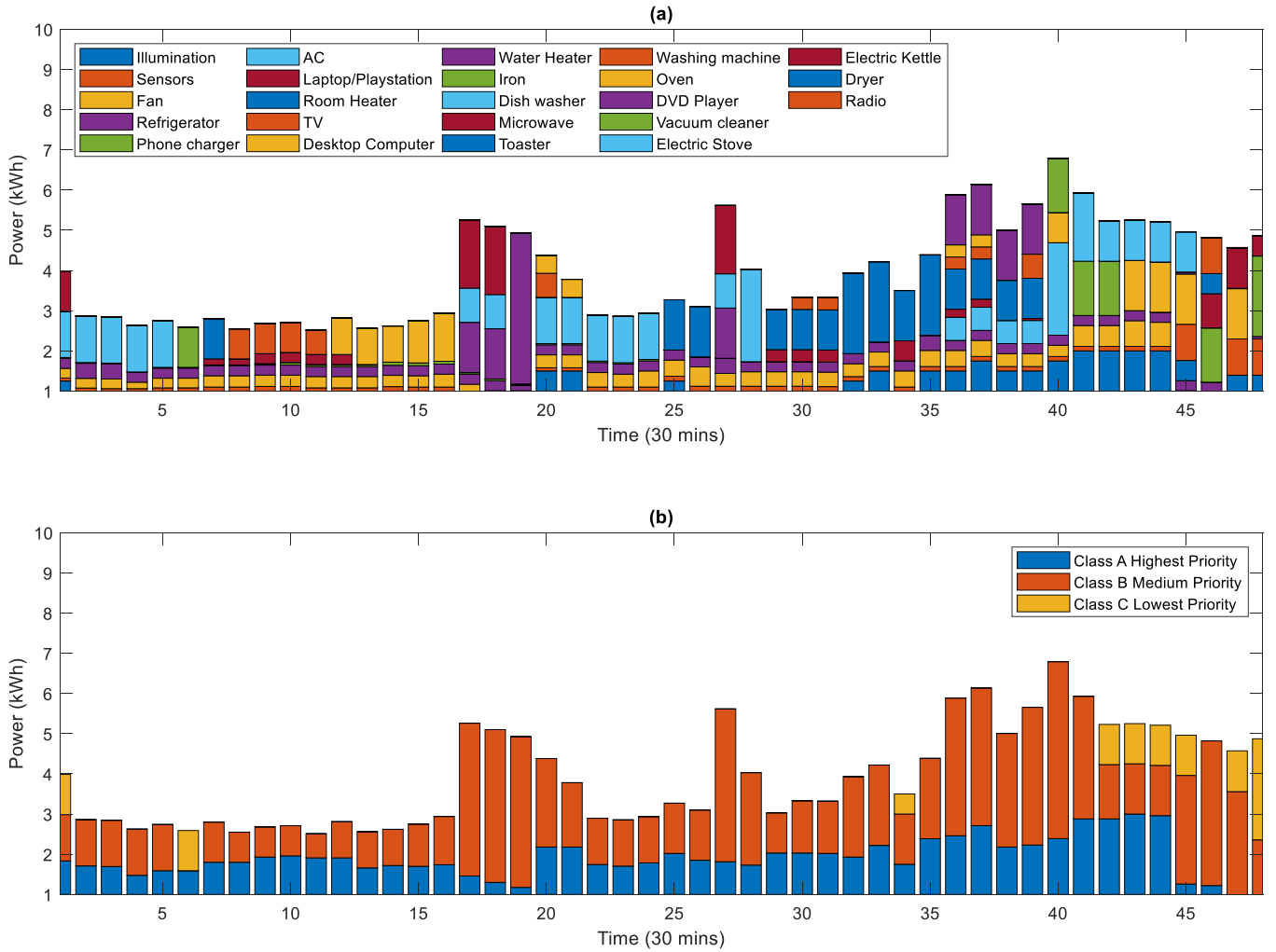


Fig. 3. a) Smart home appliances data (b) Smart home appliances data after prioritization.

5.1. Smart appliances demand data

Smart appliances in a smart home can be divided based on consumers' priorities and usage patterns (Hassan et al., 2022). We have categorized appliances into three groups. The highest priority is given to class A, which consumers consider essential for their comfort based on their preferences, and its significance cannot be neglected. Class B represents medium-priority loads, comprising appliances that can be deferred to later intervals or turned off in response to input signals, such as high prices. Lastly, class C includes the lowest priority appliances, which can be neglected, or not using them does not disrupt the regular operation. Table 1 provides smart appliances' power rating, priority level, and time of use (ToU). Each smart appliance can be represented by a set denoted by A_n for $n = 1, 2, \dots, N$, B_n for $n = 1, 2, \dots, N$ and C_n for $n = 1, 2, \dots, N$. These sets comprise all the smart appliances falling under each category. The following equations provide the total energy consumed by each class of appliances within each set during each time interval t .

$$E_{j(A_t)} = E_{A_{1,t}} + E_{A_{2,t}} + \dots + E_{A_{N,t}} = \sum_t \sum_{n=1}^N E_{A_{n,t}} \quad (30)$$

$$E_{j(B_t)} = E_{B_{1,t}} + E_{B_{2,t}} + \dots + E_{B_{N,t}} = \sum_t \sum_{n=1}^N E_{B_{n,t}} \quad (31)$$

$$E_{j(C_t)} = E_{C_{1,t}} + E_{C_{2,t}} + \dots + E_{C_{N,t}} = \sum_t \sum_{n=1}^N E_{C_{n,t}} \quad (32)$$

Where $\sum E_{A_{n,t}}$, $\sum E_{B_{n,t}}$, and $\sum E_{C_{n,t}}$ presents the energy consumed by class A, B, and C appliances. We can express the total energy consumed by a set of appliances in a smart home at a time t as follows:

$$E_{j(t)} = TE_t = \sum E_{A_{n,t}} + \sum E_{B_{n,t}} + \sum E_{C_{n,t}} \quad (33)$$

To determine the total energy consumed by a consumer throughout a day for all sets of appliances, we can express the above equation as follows:

$$E_{j(T)} = TE_d = \sum_{t=1}^T TE_t = \sum_{t=1}^T E_{A_{n,t}} + \sum_{t=1}^T E_{B_{n,t}} + \sum_{t=1}^T E_{C_{n,t}} \quad (34)$$

6. Experimental setup and results discussion

This section presents the experimental results and findings of the proposed method for energy trading in the P2P market. It considers load prioritization in smart homes to ensure the optimal use of surplus energy produced by DERs. The simulations are performed considering five consumers and seven prosumers. To ensure a balanced demand and supply of electricity, we have also considered the main grid in our analysis, which allows the import or export of energy in the case of an imbalance between supply and demand. Prosumers have surplus energy they want to trade into the P2P market after self-consumption, while consumers are interested in buying energy from the P2P market or the grid.

The data for prosumers' surplus energy, proposed prices, consumers' demand, offered prices, and grid import/export prices are taken from

Table 1
Summary of loads.

Name	Priority	Power Rating (kWh)	Time of operation	ToU of appliances T = 30 min
Illumination	A	0.5	48	48
Laptop	A	0.1	48	12
Sensors	A	0.01	48	48
Fan*	A	0.080	48	48
Phone charger	A	0.010	48	24
Refrigerator	A	0.25	48	48
Dishwasher	B	1.7	24	2
Washing Machine	B	1.8	24	4
Dryer	B	2.5	24	2
TV	B	0.3	24	10
Desktop	B	0.3	24	10
Oven	B	2.5	24	4
Iron	B	2.7	24	4
Microwave	B	1.7	24	2
AC	B	1.15	24	16
Toaster	B	1	24	4
Heater*	B	2	24	12
Water heater*	B	2.5	24	8
DVD Player	B	0.025	24	4
Vacuum Cleaner	C	2	12	2
Radio	C	0.015	12	2
Stove	C	2	12	2
Electric Kettle	C	1	12	2

Seasonal Load*

(Amin et al., 2020), which provides data sampled over a 30-minute time interval. The data for smart home appliances are taken from (Melhem et al., 2017; Longe et al., 2015).

To demonstrate the effectiveness of the proposed method, we compare the results in terms of energy cost, i.e., energy bill, with the grid and previous work (Amin et al., 2020; Tushar, 2019). Furthermore, we evaluate the revenue generated for prosumers and compare it against outcomes from the main grid and the previous methodology (Kanakadhurga and Prabakaran, 2021). Additionally, we compare the SI of consumers with the approach introduced in (Kanakadhurga and Prabakaran, 2021).

The authors in (Amin et al., 2020; Tushar, 2019) introduce game-theoretic models for P2P energy trading among peers in the P2P energy market under mid-market pricing. They do not consider the potential of smart appliances in smart homes with the available energy from DERs. Furthermore, the authors in (Kanakadhurga and Prabakaran, 2021) proposed a DR-based P2P energy trading system where participants can trade energy to the P2P community based on real-time prices (RTP). However, this system forces prosumers to sell their surplus energy at the declared RTP, which may lead to dissatisfaction among some participants and no competition.

7. Evaluation of peers' social welfare

We did not consider the energy storage system at the prosumer end during the analysis. This omission is primarily due to the high battery and associated degradation costs (Yi et al., 2001). Firstly, based on the SDR, the proposed model allocates the energy to consumers in the form of different pairs described in Algorithm 1. A consumer can buy energy from different prosumers based on its demand. Secondly, suppose the surplus energy is less than the demand. In that case, the model first evaluates the energy deficit time interval w.r.t demand, prioritizes loads of different consumers based on δ , and provides energy to the energy-conscious consumers from the grid, and updates the demand curve at each time interval. The rest of the consumers receive energy for the prioritized loads. Based on this information, P2P MO allocates energy to the price-conscious consumers at the current time interval or finds the nearest time interval with surplus energy and reallocates these consumers to this time interval using Algorithm 2, following an exhaustive search algorithm. Finally, peers can trade energy with the grid in a supply-demand imbalance.

A random time interval, t , is chosen to analyze the results of the proposed energy trading method. The data at this time interval is given below:

$$E.D_s = \{7.677, 0.419, 4.954, 0.6007, 2.74\} \quad (35)$$

$$\delta = \{0.8, 0.1, 0.3, 1, 0.8\} \quad (36)$$

$$E.S_s = \{2.632, 1.854, 3.843, 1.078, 2.643, 0.906, 3.712\} \quad (37)$$

The data in Eqs. 35 and 37 show consumers' and prosumers' demand and surplus energy set at a time interval of t . In addition, the data in Eq. 36 describe the willingness of consumers to reallocate energy for low-priority loads. The total demand of consumers is 16.391 kWh, while the available surplus of prosumers is 16.672 kWh, with 0.281 kWh of remaining surplus energy. Consumers and prosumers use the criteria shown in Eqs. 19 and 20 to submit the information about proposed and offered prices.

Algorithm 1 serves as the basis for contract formulation by the P2P MO. The first contract is formed between C_4 and P_3 , where P_3 having a surplus energy of 3.84 kWh exceeds the demand of C_4 i.e., 0.6007 kWh. As a result, P_3 has a remaining surplus of 3.23 kWh after selling it to C_4 , and the energy cost at time t for C_4 is 2.555 cents. Later, this prosumer gets engaged with the following consumer in the list, i.e., C_5 . The demand for C_5 is 2.74 kWh, while P_3 has a surplus of 3.23 kWh remaining. As a result, the next contract will be formed between these two peers. The energy cost of C_5 during this time interval is 11.699 cents. Moving on, the next consumer on the list is C_1 with a demand of 7.677 kWh. However, the surplus energy of P_3 is only 0.49 kWh, which is insufficient to meet C_1 demand. Consequently, C_1 will send trading requests to the next prosumers on the list: i.e., P_7, P_1 and P_5 . After purchasing energy from these prosumers, the trading cost for C_1 amounts to 32.679 cents. After this transaction, the remaining surplus energy of P_5 is 1.8018 kWh, which will be allocated to the subsequent consumer C_3 with a demand of 4.954 kWh. Since this prosumer alone cannot satisfy C_3 demand, the following prosumers in the trading list are P_2, P_4 and P_6 . With contributions from these prosumers, the total energy cost for C_3 is 20.924 cents. After this exchange, P_6 retains a surplus of 0.6823 kWh, which will be directed to the final consumer in the list, C_2 who has a demand of 0.429 kWh. The energy bill for C_2 in this time interval amounts to 1.8018 cents.

Table 2

Lists of consumers and prosumers.

Consumer ID	Consumers' eagerness (δ)	Consumers' List	Prosumers ID	Prosumers' Reputation (λ)	Prosumers' List
C_1	0.8	$\{C_4, C_5, C_1, C_3, C_2\}$	P_1	0.661	$\{P_3, P_7, P_1, P_5, P_2, P_4, P_6\}$
C_2	0.1	$\{C_4, C_5, C_1, C_3, C_2\}$	P_2	0.443	$\{P_3, P_7, P_1, P_5, P_2, P_4, P_6\}$
C_3	0.3	$\{C_5, C_1, C_4, C_3, C_2\}$	P_3	0.951	$\{P_3, P_7, P_1, P_5, P_2, P_4, P_6\}$
C_4	1	$\{C_4, C_5, C_1, C_3, C_2\}$	P_4	0.261	$\{P_3, P_7, P_1, P_5, P_2, P_4, P_6\}$
C_5	0.8	$\{C_5, C_1, C_4, C_3, C_2\}$	P_5	0.657	$\{P_3, P_7, P_1, P_5, P_2, P_4, P_6\}$
–	–	–	P_6	0.220	$\{P_3, P_7, P_1, P_5, P_2, P_4, P_6\}$
–	–	–	P_7	0.911	$\{P_3, P_7, P_1, P_5, P_2, P_4, P_6\}$

Table 3

Contract formulation details.

Consumer ID	$\hat{C}_{C \leftrightarrow P}$ (Contract)	Demand (kWh)	Surplus (kWh)	Remaining Demand (kWh)	Remaining Surplus (kWh)	Pairwise Trading Price (T_p)
C_1	$C_1 \leftrightarrow P_3, P_7, P_1, P_5$	7.778	9.483	0	1.7054	{4.262, 4.278, 4.232, 4.253}
C_2	$C_2 \leftrightarrow P_6$	0.429	0.6823	0	0.2533	{4.2}
C_3	$C_3 \leftrightarrow P_5, P_2, P_4, P_6$	4.954	5.6364	0	0.6823	{4.178, 4.259, 4.232, 4.272}
C_4	$C_4 \leftrightarrow P_3$	0.6008	3.84	0	3.2392	{4.259}
C_5	$C_5 \leftrightarrow P_3$	2.74	3.2392	0	0.4992	{4.270}

The trading price between each pair is calculated using Eq. 22. Table 2 provides a list of consumers and prosumers who are ready for energy trading, while the details of the contract between these peers are illustrated in Table 3.

In a P2P energy trading system, imbalances between the demand and supply of energy can arise among prosumers and consumers. To bridge this gap, the P2P MO assesses the value of δ along with the corresponding prices from each consumer and prosumer. Based on this information, the P2P MO establishes contracts that differentiate between consumers considering their δ values. These contracts enable prosumers with higher λ values to sell it to consumers with higher δ values and offered prices. This approach ensures that consumers willing to reallocate their non-essential load are prioritized and supplied with energy from the P2P market. A random time interval is selected to evaluate the outcomes of the proposed energy trading system during an overloaded time interval. The data for this time interval, used to explain the contract formulation, is presented in the equations below:

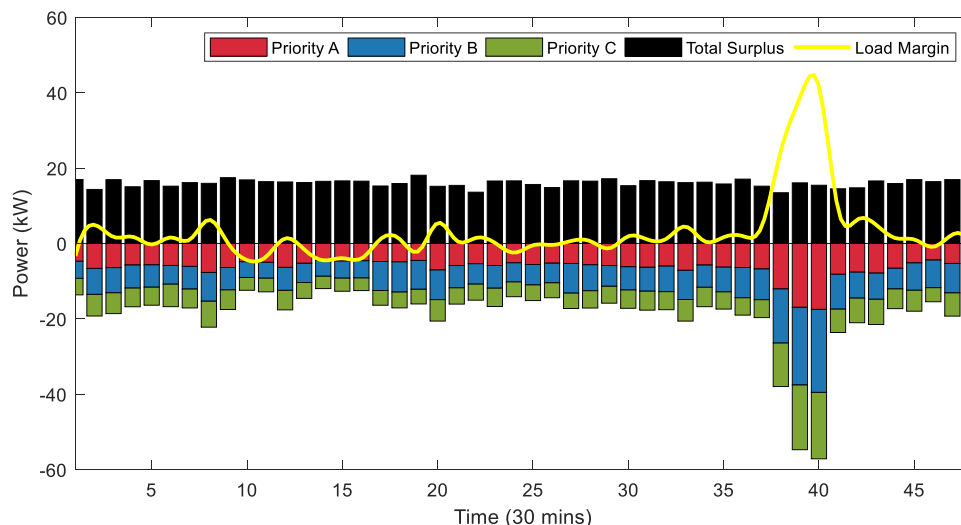
$$E.D_s = \{9.765, 0.421, 4.954, 0.645, 2.845\} \quad (38)$$

$$\delta = \{0.6, 0.6, 0.9, 0.1, 0.1\} \quad (39)$$

$$E.S_s = \{2.126, 1.957, 3.553, 1.460, 2.747, 0.401, 4.675\} \quad (40)$$

The data provided in Eqs. 38 and 40 illustrates the energy demand and surplus within the P2P community for consumers and prosumers, respectively. Eq. 39 details consumers' willingness. The information reveals that the total energy demand of peers amounts to 18.63 kWh, which exceeds the available surplus of 16.8 kWh. It is evident that not all consumers in the P2P market can be supplied with energy. To address this issue, the system examines the δ values and prioritized loads using Algorithm 2 to differentiate and give priority to consumers. Based on the information from the smart meters, the method reallocates their loads and supplies them with energy from the P2P market either at the time t_s or t_r following the provision of energy to energy-conscious consumers from the grid. The proposed system is designed to efficiently manage energy allocation by identifying overloaded time intervals, analyzing load margin, and implementing the prioritization technique. Initially, the system identifies overloaded time intervals and employs the load margin to quantify the excess load. The central control unit operates and receives a priority data signal from the remote control unit. Fig. 4 illustrates the load margin and available surplus energy with the priority load data. Based on this information, the system distinguishes between the loads and determines priority access.

In addition, the system identifies overloaded time intervals and initiates load reallocation. Fig. 5(a) shows the overload intervals where the

**Fig. 4.** Load margin analysis and priority data.

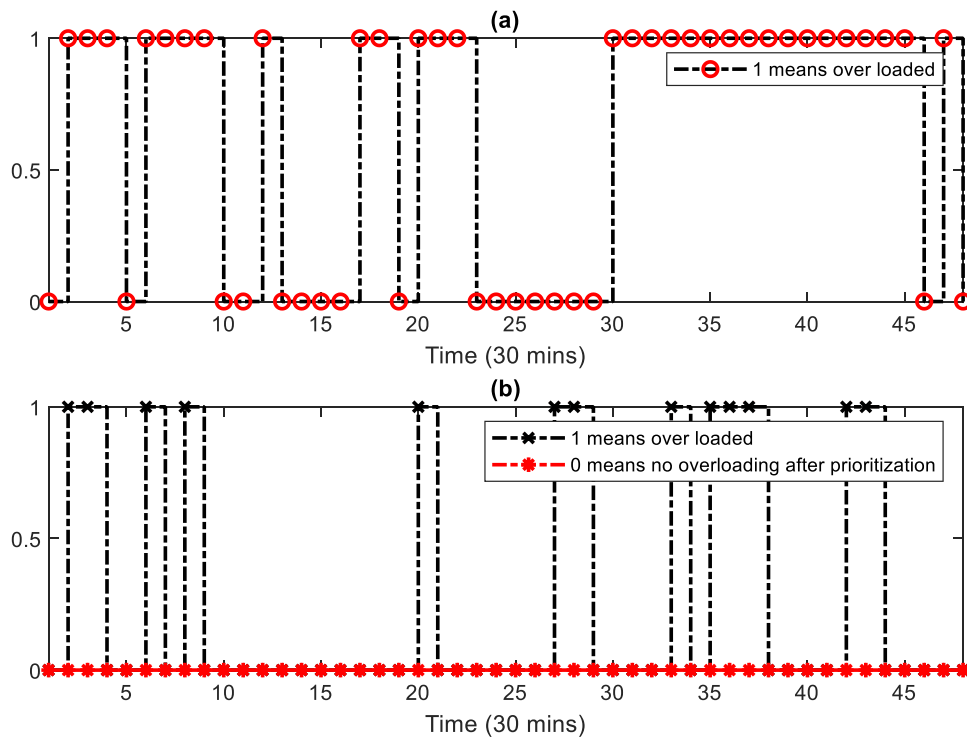


Fig. 5. (a) Time intervals ($T.D_i > T.S_i$) (b) Energy provided from surplus energy intervals after appliance prioritization and reallocation.

energy is insufficient to supply all the loads. Fig. 5(b) indicates the identification of energy deficit time intervals following load margin analysis. Then, the system implements the prioritization technique to reallocate lower-priority loads for price-conscious consumers. The load reallocation strategy first prioritizes the consumers' load, checks the information received from the smart meters, and supplies energy to the energy-conscious consumers from the grid. Then, it reallocates some portion of the load (priority B, C, or both) to time intervals with surplus energy. The grid is considered the last option. The system ensures maximum use of green energy through proper coordination between peers and P2P MO.

Table 4 lists consumers and prosumers willing to participate in energy trading at the selected time interval. Consumers who reported lower willingness are considered energy-conscious and will buy energy from the grid or the P2P market at any price. Consequently, at this time interval, they will buy energy from the grid at the grid import price, i.e., 5.221. To give priority access to the P2P market, we accommodate consumers who have reported a high δ value for purchasing energy. After prioritizing and verifying the loads of each consumer using Algorithm 2, the system sends the data back to the P2P MO and formulates contracts between the peers. The data shows that the δ reported by C_4 and C_5 is 0.1, which shows they do not want to change their load pattern and at this interval, the energy demand is higher than the available surplus. These consumers will buy energy from the grid, and the contract

will charge additional fees for delivering energy to consumers who reported higher δ to avoid false reporting. The total energy cost for C_4 and C_5 at this time interval is 3.36 and 14.82 cents.

The energy demand for C_3 amounts to 4.9594 kWh, and the prosumers in the trading list are P_7 and P_3 . After selling energy to this consumer, the remaining energy of P_3 is 3.274 kWh. The energy cost for C_3 during this time interval is 20.95 cents. The subsequent consumer in the list is C_1 , with a demand of 9.765 kWh. Based on the consumers' δ value, it indicates a willingness to purchase for the load priorities A and B from the P2P market while reallocating the load of priority C, which holds the lowest priority. However, as mentioned above, after supplying energy to the energy-conscious consumers from the grid, if the energy available for the prosumers is sufficient to meet consumers' demand, then the consumers will buy energy at the current time interval. This consumer forms a trading pair with P_3, P_5 and P_1 , resulting in a total energy cost of 44.83 cents. Following the sale of energy to this consumer, the remaining energy of P_1 is 1.802 kWh. The subsequent consumer in the list i.e., C_2 has a demand of 0.4216 kWh, and it buys energy from P_1 . The overall energy cost for C_2 during this time interval totals 1.74 cents.

The details of the contract between these peers are shown in Table 5. Fig. 6(a–e) shows the consumers' energy bills and compares their costs with the cost of imported energy from the main grid and the method proposed in (Amin et al., 2020; Tushar, 2019). Similarly, Fig. 7(a–g) shows the prosumers' revenue, and compares it when the prosumers sell

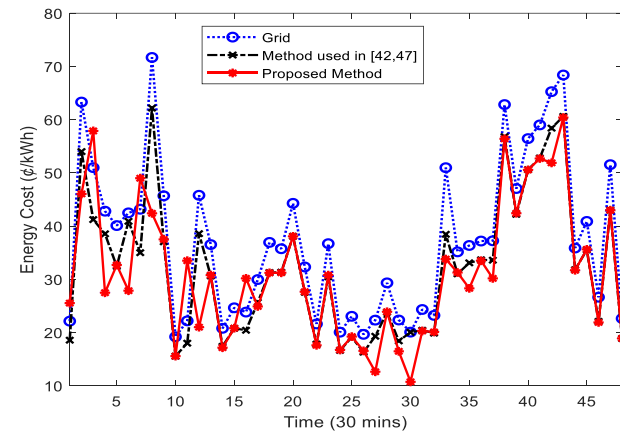
Table 4
Lists of consumers and prosumers.

Consumer ID	Consumers' eagerness (δ)	Consumers' List	Prosumers' ID	Prosumers' Reputation (λ)	Prosumers' List
C_1	0.6	$\{C_3, C_1, C_2, C_4, C_5\}$	P_1	0.532	$\{P_7, P_3, P_5, P_1, P_2, P_4, P_6\}$
C_2	0.6	$\{C_4, C_5, C_1, C_3, C_2\}$	P_2	0.466	$\{P_7, P_3, P_5, P_1, P_2, P_4, P_6\}$
C_3	0.9	$\{C_3, C_1, C_2, C_4, C_5\}$	P_3	0.866	$\{P_7, P_3, P_5, P_1, P_2, P_4, P_6\}$
C_4	0.1	$\{C_4, C_5, C_1, C_3, C_2\}$	P_4	0.353	$\{P_7, P_3, P_5, P_1, P_2, P_4, P_6\}$
C_5	0.1	$\{C_3, C_1, C_2, C_4, C_5\}$	P_5	0.700	$\{P_7, P_3, P_5, P_1, P_2, P_4, P_6\}$
–	–	–	P_6	0.097	$\{P_7, P_3, P_5, P_1, P_2, P_4, P_6\}$
–	–	–	P_7	1.132	$\{P_7, P_3, P_5, P_1, P_2, P_4, P_6\}$

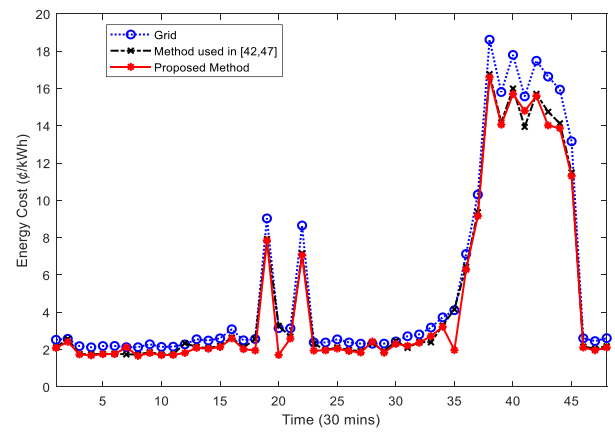
Table 5

Contract formulation details.

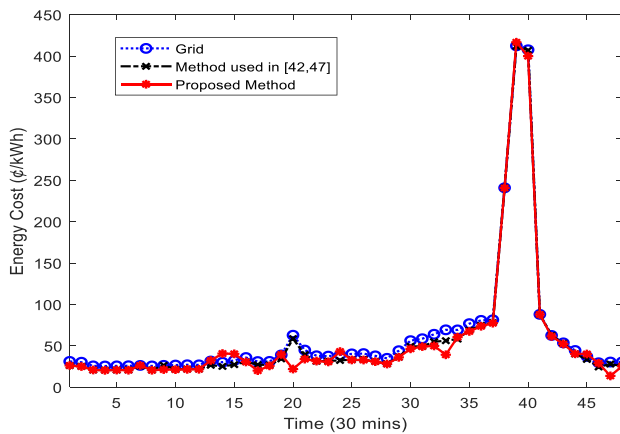
Consumer ID	$\hat{C}_{C \rightarrow P}$ (Contract)	Demand (kWh)	Surplus (kWh)	Remaining Demand (kWh)	Remaining Surplus (kWh)	Pairwise Trading Price (T_p)
C_1	$C_1 \leftrightarrow P_3, P_5, P_1$	9.765	9.483	0	1.802	{4.295, 4.206, 4.240}
C_2	$C_2 \leftrightarrow P_1$	0.4216	0.9067	0	5.19	{4.134}
C_3	$C_3 \leftrightarrow P_7, P_3$	4.954	8.228	0	3.274	{4.23, 4.215}
C_4	$C_4 \leftrightarrow G$	0.6454	–	0	3.2392	{5.221}
C_5	$C_5 \leftrightarrow G$	2.84	–	0	0.4992	{5.221}



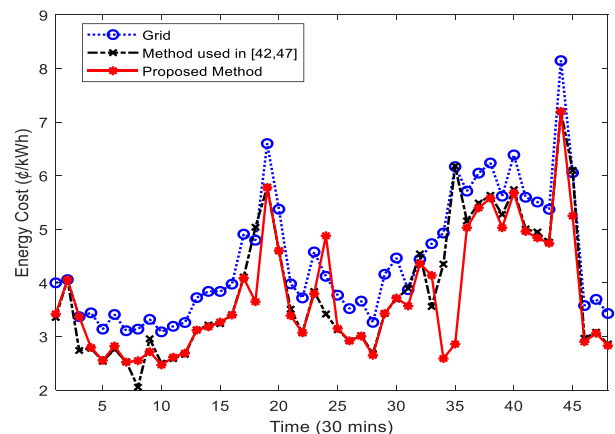
(a)



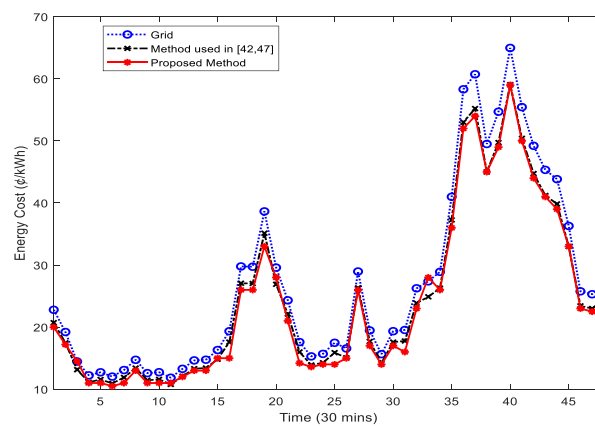
(b)



(c)

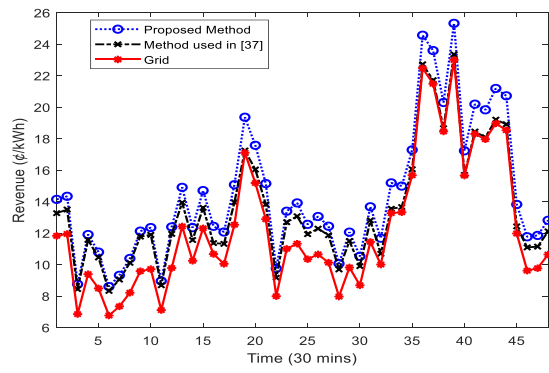


(d)

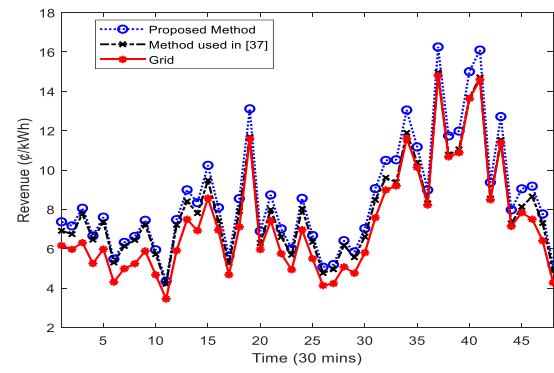


(e)

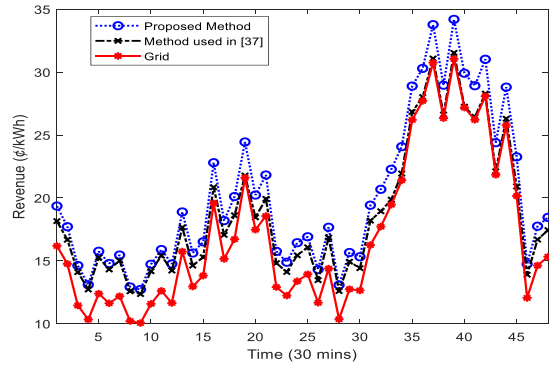
Fig. 6. (a) C_1 Energy bill (b) C_2 Energy bill (c) C_3 Energy bill (d) C_4 Energy bill (e) C_5 Energy bill.



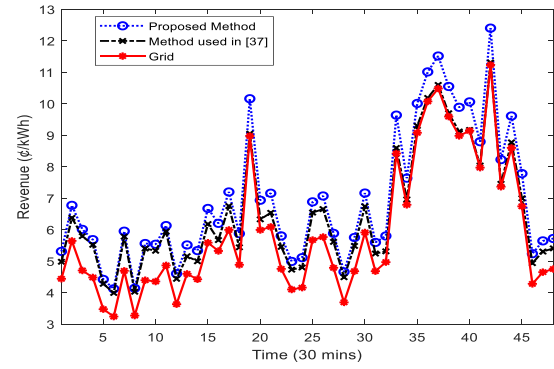
(a)



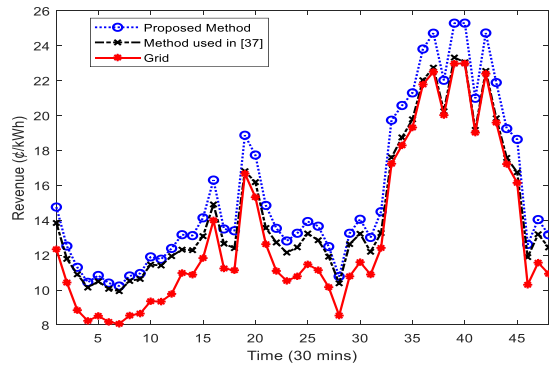
(b)



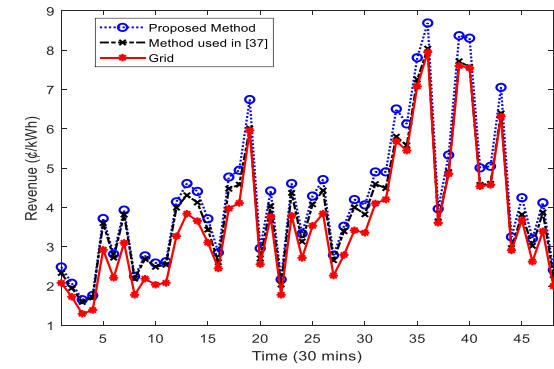
(c)



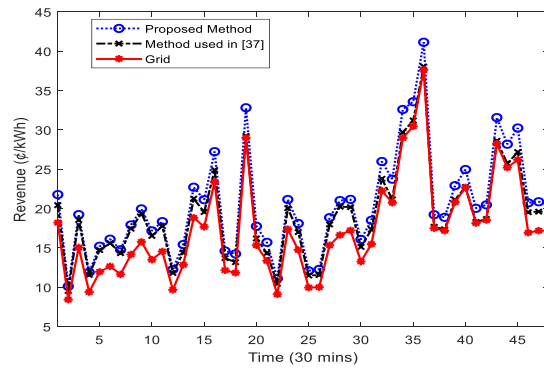
(d)



(e)



(f)



(g)

Fig. 7. (a) P_1 Revenue (b) P_2 Revenue (c) P_3 Revenue (d) P_4 Revenue (e) P_5 Revenue (f) P_6 Revenue (g) P_7 Revenue.

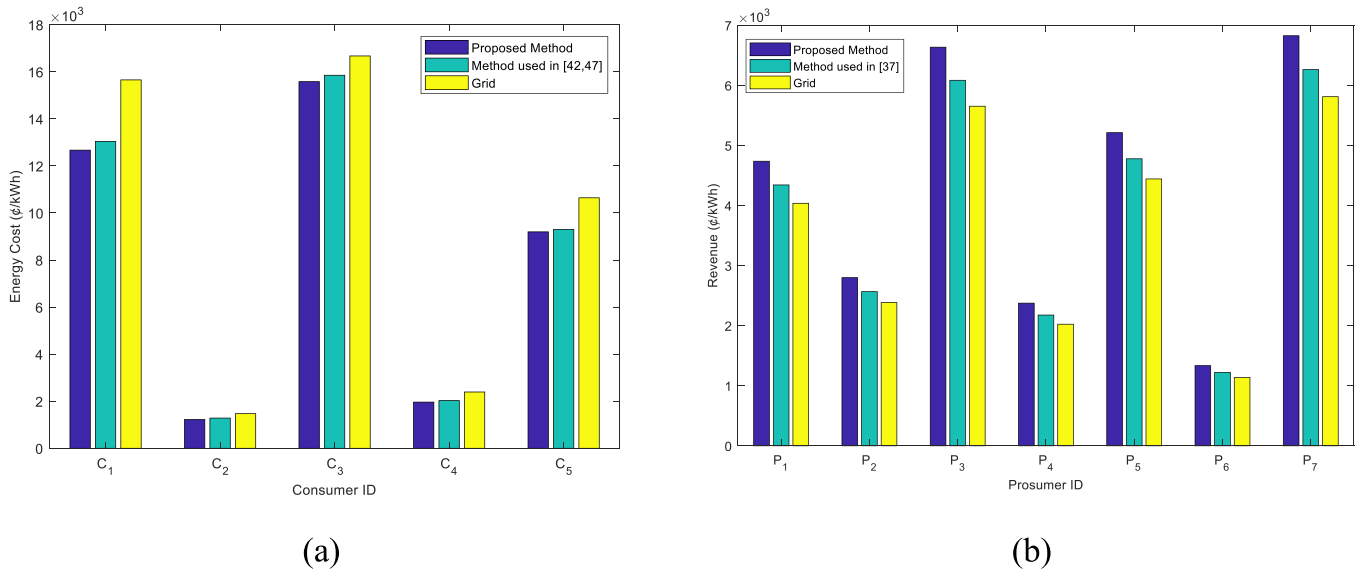


Fig. 8. (a) Consumers' energy bill comparison (b) Prosumers' revenue comparison.

to the main grid and the method proposed in (Kanakadhurga and Prabakaran, 2021). Simulation results for consumers and prosumers over one week are shown in Fig. 8(a,b). Fig. 8(a) compares the consumers' total energy cost of the proposed method with other methods when consumers buy energy from the grid and with the study presented in (Amin et al., 2020; Tushar, 2019). On the other hand, Fig. 8(b) compares the revenue earned by the prosumers using the proposed method with other methods when they sell energy to the grid, as well as with the study presented in (Kanakadhurga and Prabakaran, 2021).

The results show significant savings in consumers' energy bills and the potential for prosumers to maximize their revenues by trading energy with consumers. The proposed energy trading system between smart homes in a P2P community uses the DSM technique to optimize energy consumption and trading in the P2P energy market.

To further illustrate the benefits of the proposed energy trading system, Tables 6 and 7 show the percentage decrease in consumers' energy costs and the increase in prosumers' revenues over one week, respectively. The net energy bill and revenue columns in the tables represent consumers' energy bills and the revenue earned by prosumers in the P2P energy market. Fig. 8(a, b), along with Tables 6 and 7,

demonstrate that the proposed energy trading system has the potential to decrease consumers' energy bills while simultaneously increasing prosumers' revenues. The system achieves this by prioritizing consumer load and allocating energy from the surplus energy time intervals through optimal management and utilization of the energy generated by DERs. Fig. 5(a,b) shows the results of DSM and load prioritization through which reallocation is performed to provide energy to the lower priority.

After identifying energy deficit intervals, prioritization and reallocation is performed based on consumers' δ . The results are compared with the work proposed in (Kanakadhurga and Prabakaran, 2021) and (Amin et al., 2020; Tushar, 2019) to validate the effectiveness of the proposed approach. The results show that the proposed method outperforms these works, as shown in Fig. 8(a,b).

In addition, the proposed method reduces the amount of electricity exported to the grid, which benefits prosumers when they sell surplus energy to the consumers. It reduces the amount of electricity imported from the grid, which reduces consumers' electricity bills after prioritization. In summary, the proposed energy trading system with the DSM method and load prioritization among P2P communities can optimize

Table 6
Consumers' energy bills comparison.

Consumer ID.	Consumers' Net Energy Cost (€)	Consumers' Energy Cost (€)(Amin et al., 2020; Tushar, 2019)	Consumers' Energy Cost (€) Grid	%age Reduction w.r.t Grid	%age Reduction w.r.t(Amin et al., 2020; Tushar, 2019)
C ₁	126,70.44	130,43.11	156,56.16	19.07	2.86
C ₂	1223.355	1283.839	1471.231	16.85	4.71
C ₃	155,81.24	158,52.59	166,73.11	6.55	1.71
C ₄	1962.831	2026.484	2391.984	17.94	3.14
C ₅	9197.472	9302.010	106,45.20	13.6	1.12

Table 7
Prosumers' revenue comparison.

Prosumer ID.	Prosumers' Net Revenue (€)	Prosumers' Revenue (€)(Kanakadhurga and Prabakaran, 2021)	Prosumers' Revenue Grid	%age increase w.r.t Grid	%age increase w.r.t(Kanakadhurga and Prabakaran, 2021)
P ₁	4736.77	4342.22	4036.684	17.34	9.09
P ₂	2799.26	2565.77	2385.445	17.35	9.1
P ₃	6634.96	6084.11	5652.346	17.38	9.05
P ₄	2374.72	2176.61	2024.154	17.32	9.1
P ₅	5211.82	4777.18	4441.935	17.33	9.1
P ₆	1334.19	1222.85	1137.316	17.31	9.1
P ₇	6825.05	6261.79	5810.430	17.46	9

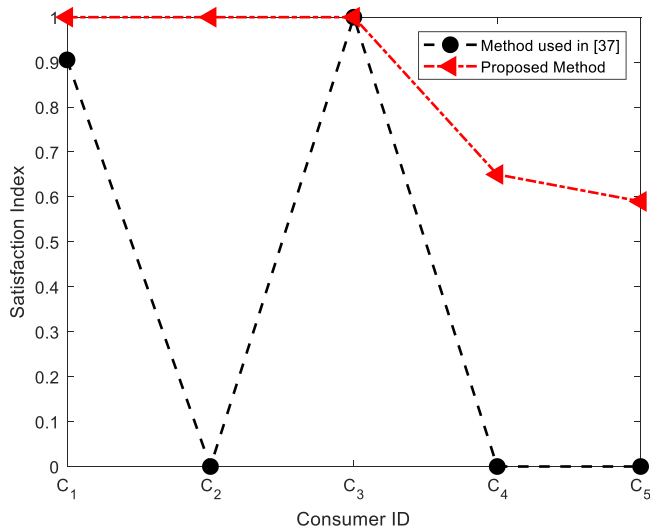


Fig. 9. Comparison of consumers' satisfaction index.

energy use and trading in the P2P energy market, benefiting consumers and prosumers by optimally using available energy from DERs.

7.1. Evaluation of consumers' satisfaction index (SI)

Finally, to evaluate the SI of the consumers using the proposed method, we have chosen a random time interval. From Fig. 9, it is evident that the SI of the consumers is increased compared to the work presented in (Kanakadhurga and Prabakaran, 2021). The illustration showcases that by implementing the proposed energy allocation policy, the satisfaction level of C₃ remains consistent with that of (Kanakadhurga and Prabakaran, 2021).

On the other hand, an increase in SI values ranging from 10 % to 100 % is observed among the other consumers, i.e., 10 % for C₁, 100 % for C₂, 65 % for C₄, and 59 % for C₅, compared to the study in (Kanakadhurga and Prabakaran, 2021) after prioritizing the higher priority loads and allocating energy from the P2P market. Table 8 outlines all consumers' overall percentage change in SI relative to the (Kanakadhurga and Prabakaran, 2021). Overall, on average, the SI of the consumers is increased by 46.8 %. These results highlight the effectiveness of the proposed method in improving consumers' SI.

7.2. Strengths and limitations of this study

In this study, we present a novel approach for managing DERs through P2P energy trading within a P2P community of smart homes, and we propose an optimized energy allocation policy. The proposed method maximizes DERs utilization by efficiently coordinating energy generation and consumption. Notably, our approach revolves around a priority-based scheme in which smart home appliances are classified into three distinct classes: class A, class B, and class C, with the highest priority assigned to class A appliances.

One of the primary strengths of our approach lies in managing DERs through P2P energy trading with load prioritization among smart homes. By accommodating the stochastic nature of DERs, our framework ensures effective energy provisioning for consumers, fostering mutually beneficial interactions within the community. Moreover, the approach reinforces consumers' self-sufficiency and enhances their SI by reducing grid dependency. Furthermore, our strategy contributes to the enhancement of overall social welfare. Maximizing energy supply duration to higher-priority loads aligns energy distribution with user needs, enhancing the system's effectiveness.

However, there are certain limitations in this study that could be addressed in future research work. Firstly, our focus primarily remained on consumer-side appliance behavior, neglecting the perspective of prosumers within the P2P energy trading framework. We assumed that prosumers possess surplus energy for trading after self-consumption. Integrating prosumers' appliances behavior and decision-making could substantially enhance the efficiency and robustness of the P2P energy system. Secondly, the assignment of priority levels to smart home appliances was based on daily usage patterns.

Nevertheless, this approach might not account for individual variations arising from diverse lifestyles and work schedules. To address this limitation, a potential avenue for improvement involves clustering consumers based on usage behavior, enabling the tailored optimization of appliance priorities within each cluster, leading to more effective utilization of DERs while accurately catering to consumers' distinctive requirements. Lastly, our study omits the consideration of weather-dependent energy generation from DERs under diverse weather conditions. To refine our model, future studies could integrate this parameter and construct a more precise representation by modeling prosumer generation risks. This enhancement would result in a more accurate estimation of surplus energy availability.

In summary, while our study introduces a promising framework for P2P energy trading to manage DERs, these acknowledged limitations point to valuable directions for future research enhancements. By incorporating prosumers' perspectives, optimal appliance priority assignment, and prosumer energy generation risk under diverse weather conditions, the proposed approach can be refined to meet a more comprehensive array of energy trading scenarios and individual consumer needs, considering the surplus energy available from DERs.

8. Conclusion and future work

In this study, we have proposed an innovative energy trading mechanism that considers the available energy from DERs. We explored the significant impact of optimizing load-side responses on the overall performance of an energy trading framework within a P2P community of smart homes. Our approach incorporates a load prioritization technique considering consumers' willingness to shift their low-priority loads in a P2P market.

The proposed DSM technique can maximize social welfare and hours of energy served to peers for higher priority loads in a P2P market. This technique integrates P2P energy trading with the prioritization of smart home appliances. These appliances are categorized based on user needs, and their energy supply in the P2P market is prioritized accordingly. An optimal energy allocation policy is designed to reduce grid dependency

Table 8

Comparison of consumers' satisfaction level at t.

C _{ID}	Proposed Method	Method used in (Kanakadhurga and Prabakaran, 2021)	%age change w.r.t (Kanakadhurga and Prabakaran, 2021)
C ₁	1	0.90	10 % increase
C ₂	1	0	100 % increase
C ₃	1	1	0 % increase
C ₄	0.65	0	65 % increase
C ₅	0.59	0	59 % increase

and increase the self-sufficiency of a P2P community. We proposed a load control and reallocation algorithm strategically deployable alongside DSM applications in P2P communities. Through comprehensive simulations, we demonstrated that our proposed model empowers prosumers and consumers to maximize their social welfare. By implementing load prioritization, surplus energy from DERs is efficiently utilized, resulting in substantial energy cost reductions of 14.80–2.70 % for consumers and increased revenue of 17.35–9.07 % for prosumers compared to the grid and previously proposed studies.

Moreover, our method significantly enhances the SI of consumers, marking an average increase of 46.8 %. In summary, our DSM technique significantly enhances the success of P2P energy trading by aligning energy-related goals among peers, thereby fostering more sustainable and efficient energy consumption practices.

The potential of our proposed work can be extended in different ways. An intriguing avenue for extension involves modeling the generation risk faced by prosumers under diverse weather conditions, coupled with an analysis of their appliances' usage patterns. Furthermore, the clustering of consumers according to their distinct energy consumption behaviors—taking into account lifestyle and work schedules can increase their SI.

CRedit authorship contribution statement

Syed Adrees Ahmed: Conceptualization, Methodology, Software, Writing – original draft, Writing – review & editing, Investigation, Visualization. **Qi Huang:** Supervision, Methodology, Visualization, Writing – review & editing. **Waqas Amin:** Writing – Reviewing and Editing. **Muhammad Afzal:** Writing – Reviewing and Editing. **Fazal Hussain:** Writing – Reviewing and Editing. **Muhammad Husnain Haider:** Writing – Reviewing and Editing.

Declaration of Competing Interest

We have no conflict of interest to disclose.

Data Availability

Data will be made available on request.

Acknowledgments

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